

mne.make_forward_solution

mne.make_forward_solution(*info*, *trans*, *src*, *bem*, *meg*=True, *eeg*=True, ***, *mindist*=0.0, *ignore_ref*=False, *n_jobs*=None, *on_inside*='raise', *verbose*=None)

[\[source\]](#)

Calculate a forward solution for a subject.

Parameters:

info : `mne.Info` | [path-like](#)

The `mne.Info` object with information about the sensors and methods of measurement. If [path-like](#), it should be a `str` or `pathlib.Path` to a file with measurement information (e.g. `mne.io.Raw`).

trans : [path-like](#) | `dict` | *instance of* `Transform` | `"fsaverage"` | `None`

If str, the path to the head<->MRI transform `*-trans.fif` file produced during coregistration. Can also be `'fsaverage'` to use the built-in fsaverage transformation. If trans is None, an identity matrix is assumed.

! *Changed in version 0.19:* Support for `'fsaverage'` argument.

src : [path-like](#) | *instance of* `SourceSpaces`

Either a path to a source space file or a loaded or generated `SourceSpaces`.

bem : [path-like](#) | `ConductorModel`

Filename of the BEM (e.g., `"sample-5120-5120-5120-bem-sol.fif"`) to use, or a loaded `ConductorModel`. See `make_bem_model()` and `make_bem_solution()` to create a `mne.bem.ConductorModel`.

meg : [bool](#)

If True (default), include MEG computations.

eeg : [bool](#)

If True (default), include EEG computations.

mindist : `float`

Minimum distance of sources from inner skull surface (in mm).

ignore_ref : [bool](#)

If True, do not include reference channels in compensation. This option should be True for KIT files, since forward computation with reference channels is not currently supported.

n_jobs : `int` | `None`

The number of jobs to run in parallel. If `-1`, it is set to the number of CPU cores. Requires the `joblib` package. `None` (default) is a marker for 'unset' that will be interpreted as `n_jobs=1` (sequential execution) unless the call is performed under a `joblib.parallel_config` context manager that sets another value for `n_jobs`.

on_inside : `'raise'` | `'warn'` | `'ignore'`

What to do if MEG sensors are inside the outer skin surface. If `'raise'` (default), an error is raised. If `'warn'` or `'ignore'`, the forward solution is computed anyway and a warning is or isn't emitted, respectively. 

verbose : `bool` | `str` | `int` | `None`

Control verbosity of the logging output. If `None`, use the default verbosity level. See the [logging documentation](#) and `mne.verbose()` for details. Should only be passed as a keyword argument.

Returns:

fwd : *instance of* `Forward`

The forward solution.

See also

[convert_forward_solution](#)

Notes

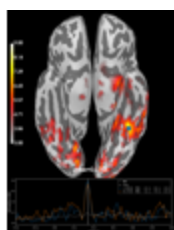
The `--grad` option from MNE-C (to compute gradients) is not implemented here. To create a fixed-orientation forward solution, use this function followed by `mne.convert_forward_solution()`.

Note

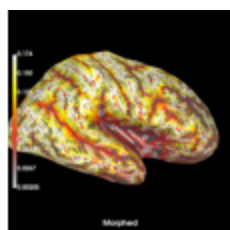
If the BEM solution was computed with [OpenMEEG](#) in `mne.make_bem_solution()`, then OpenMEEG will automatically be used to compute the forward solution.

❗ **Changed in version 1.2:** Added support for OpenMEEG-based forward solution calculations.

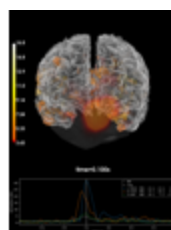
Examples using `mne.make_forward_solution`



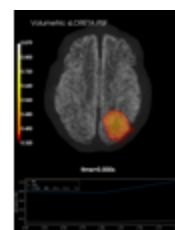
From raw data to dSPM on SPM Faces dataset



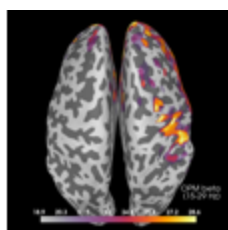
Use source space morphing



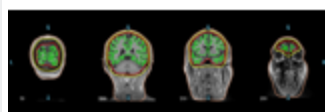
Compute MNE inverse solution on evoked data with a mixed source space



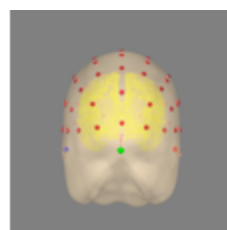
Plot point-spread functions (PSFs) for a volume



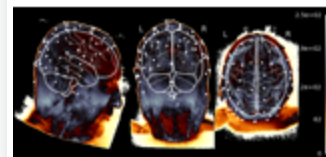
Compute source power spectral density (PSD) of VectorView and OPM data



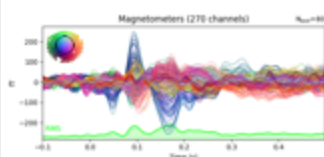
Head model and forward computation



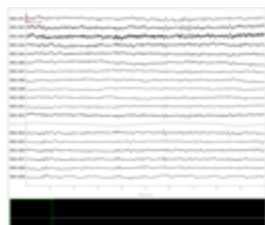
EEG forward operator with a template MRI



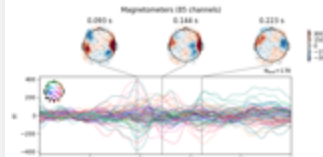
EEG source localization given electrode locations on an MRI



Working with CTF data: the Brainstorm auditory dataset



Setting the EEG reference



Preprocessing optically pumped

magnetometer
(OPM) MEG data

© Copyright 2012–2026, MNE Developers. Last updated 2026-05-07 7:11:29 GMT+3. Built with the [PyData Sphinx Theme](#) 0.17.1.